

2025 Semester 1

01211373 Machine Learning and Programming for Industry

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Lecture 01 : Course Introduction

- Course syllabus/grading policy
- Main text & materials
- Introduction to machine learning
- Python & Jupyter notebook guide



Course Syllabus

Name: Machine Learning and Programming for Industry 3(2-3-6)

Description: (from EMME curriculum) Machining learning in industry. Neural network. Hardware and Software. Perceptron. Supervised and unsupervised learning. Reinforcement learning. Loss functions. Back propagation. Binary and multiclass classification. Hyperparameters tuning. Convolutional neural networks. Transfer learning. Data pipelining. Recurrent neural networks. Sequence models. Natural language processing. Embedded device deployment.

Schedule: Mon 1 – 4 PM

Instructor: Dr Varodom Toochinda

Mobile: 084-3239613 Email: varodom.t@gmail.com

Texts:

- S. Raschka, Y. Liu and V. Mirjalili. Machine Learning with PyTorch and Scikit-Learn. Packt Publishing. 2022.
- Instructor handouts

Grading:

1) Homework Assignments	20 %	2) Project	40 %
3) Midterm	20 %	4) Final	20 %

Google Class Code : oomwxw2g

<https://classroom.google.com/c/NzY5MjM5OTg5OTcx?cjc=oomwxw2g>

Github : <https://github.com/dewdotninja/mlpi>



Course Outline (tentative)

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Lecture	Topics
1	Course introduction. Math & SW tools (Python & Jupyter notebook)
2	Perceptron. Adaline. Basics of learning.
3	Classifiers. Logistic regression. Regularization.
4	Support vector machines.
5	Decision tree learning. K-nearest neighbors.
6	Data preprocessing.
7	Dimensionality reduction. Principal component analysis.
	Midterm
8	Model evaluation. Hyperparameter tuning.
9	Regression analysis.
10	Clustering Analysis.
11	Multilayer artificial neural networks
12	Convolutional neural networks.
13	Recurrent neural networks.
14	Advanced topics. Course summary.
	Final

Main text :

- S. Raschka, Y. Liu and V. Mirjalili. Machine Learning with PyTorch and Scikit-Learn. Packt Publishing. 2022.

<https://sebastianraschka.com/blog/2022/ml-pytorch-book.html>



Software

- Python with
 - standard packages : numpy, scipy, matplotlib, panda, etc.
 - scikit-learn
 - pytorch
- Jupyter notebook
 - run on Colab
 - use docker
 - install locally
- Additional software as needed

Math

- linear algebra
 - vectors, matrices, tensors
 - eigenvalues/eigenvectors
 - singular value decomposition (SVD)
- calculus
 - derivatives, gradients, jacobians, hessians
- statistics
 - probability, stochastic process

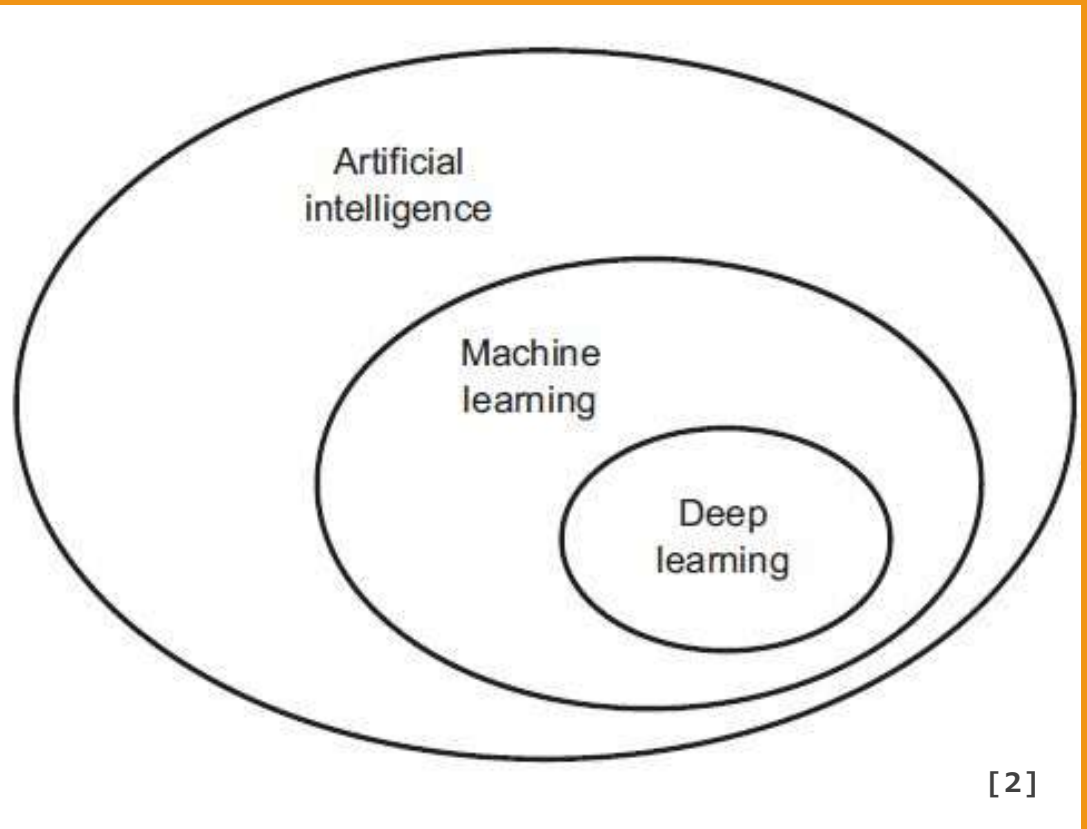
What is machine learning?

- Algorithms that analyze data and turn into knowledge
 - Detect patterns/structures
 - Classification/regression
 - Predict future events
- Subclass of AI (Artificial Intelligence)
- Trained instead of programmed

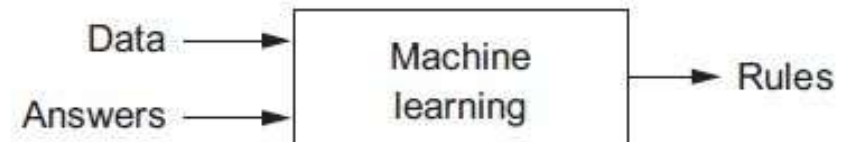
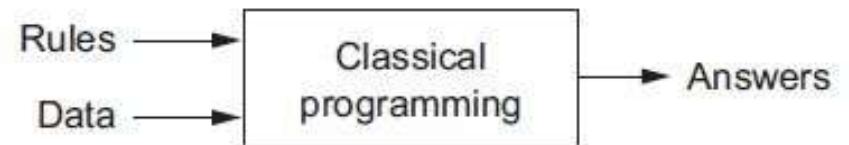
Famous quotes

- “Machine learning is the field of study that gives computers the ability to learn without being explicitly programmed.” - Arthur L. Samuel . 1959.
- “A computer program is said to learn from experience E with respect to some class of tasks T and performance measure P , if its performance at tasks in T , as measured by P , improves with experience E .” - Tom Mitchell, CMU. 1997.

AI/ML/DL



New program paradigm in ML



3 types of ML

Supervised

- labeled data (“ground truth”)
- direct feedback
- predict outcome/future

Unsupervised

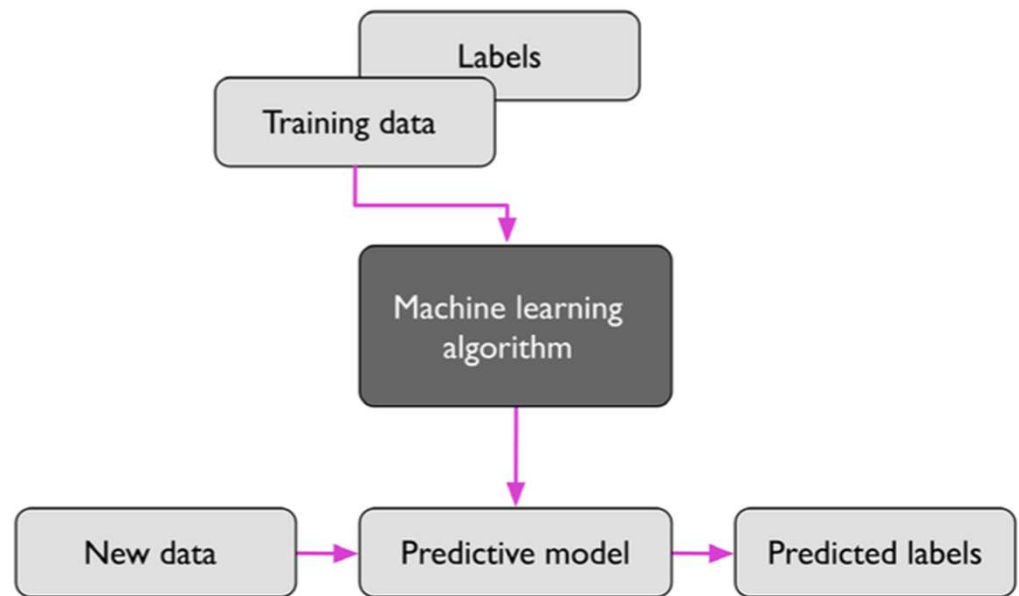
- no labels/targets
- no feedback
- find hidden structure in data

Reinforcement learning (RL)

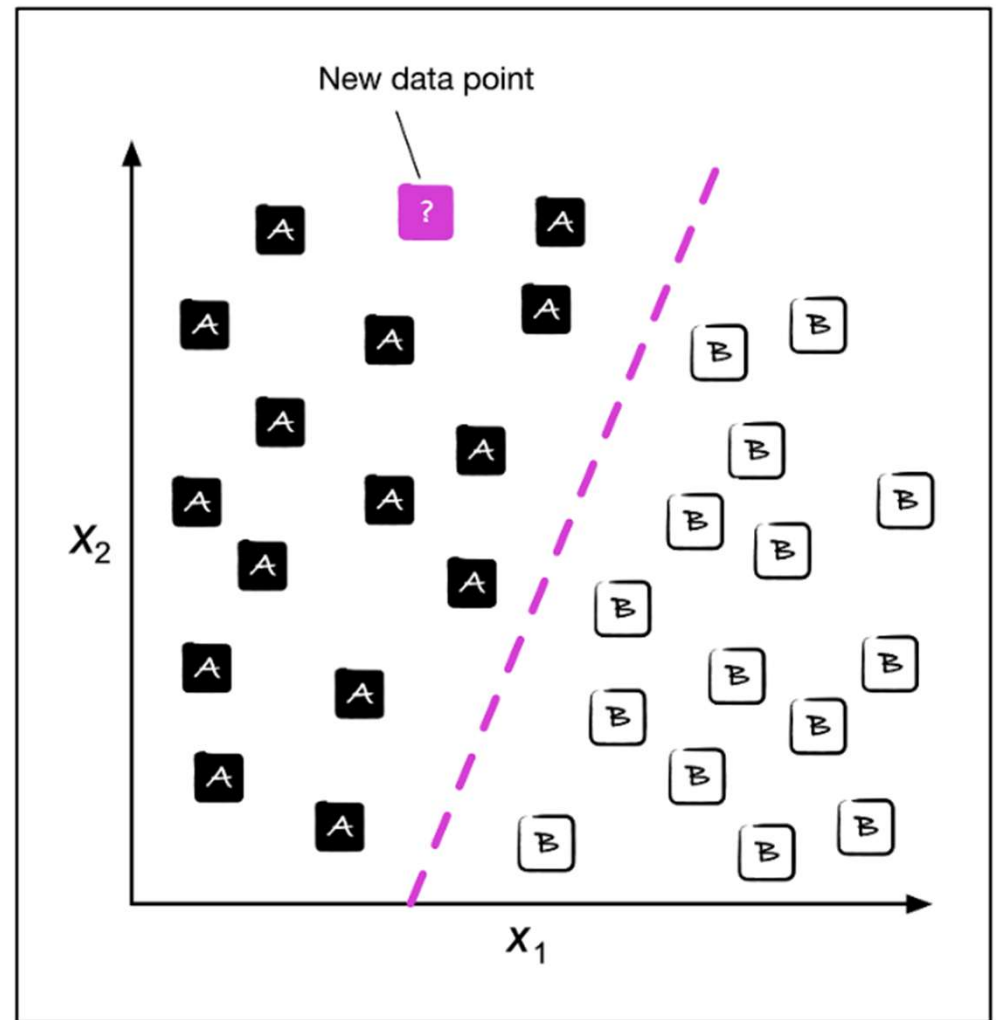
- decision process
- reward system
- learn series of action

Supervised learning

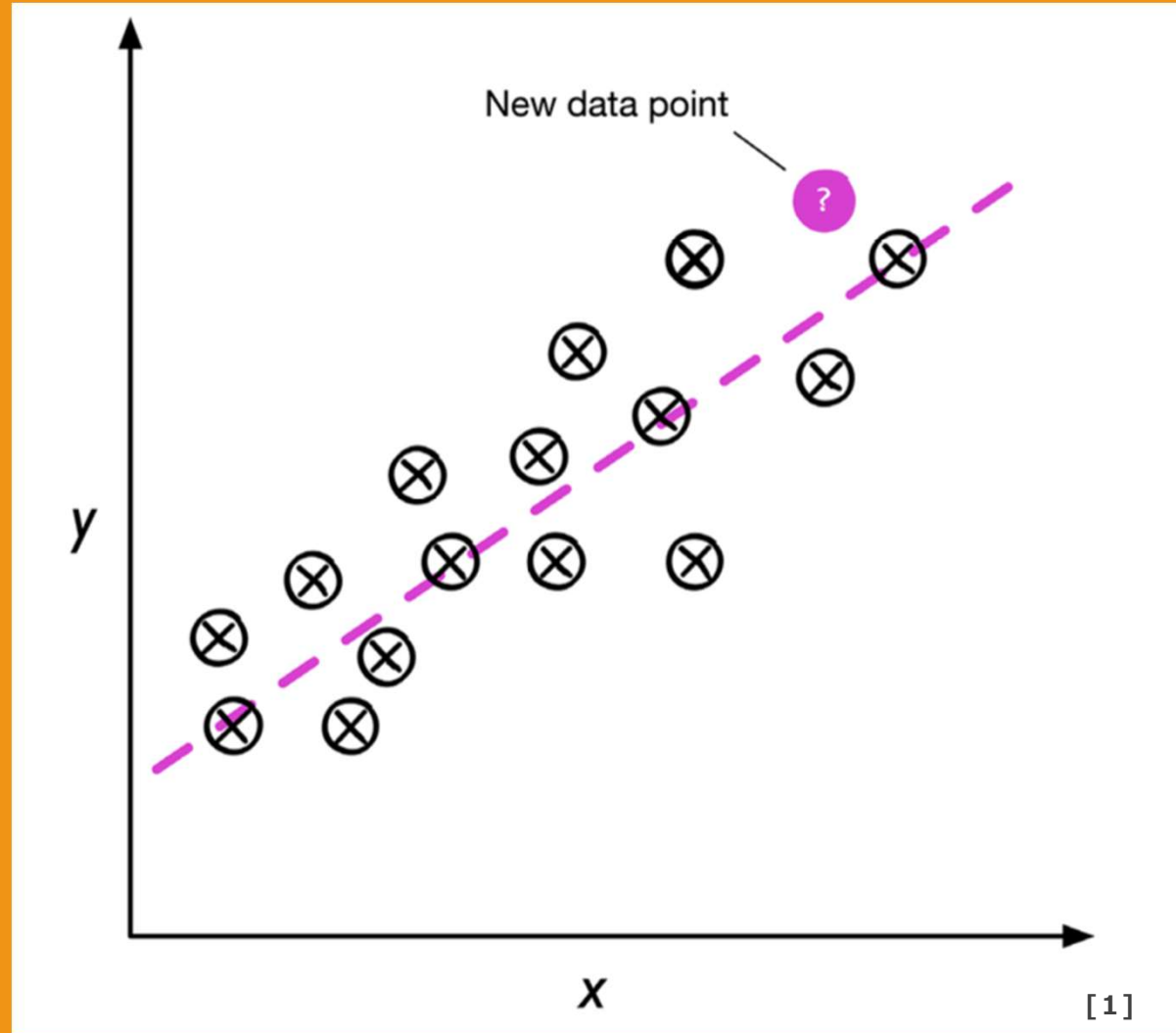
Supervised learning process



Classifying a new data point

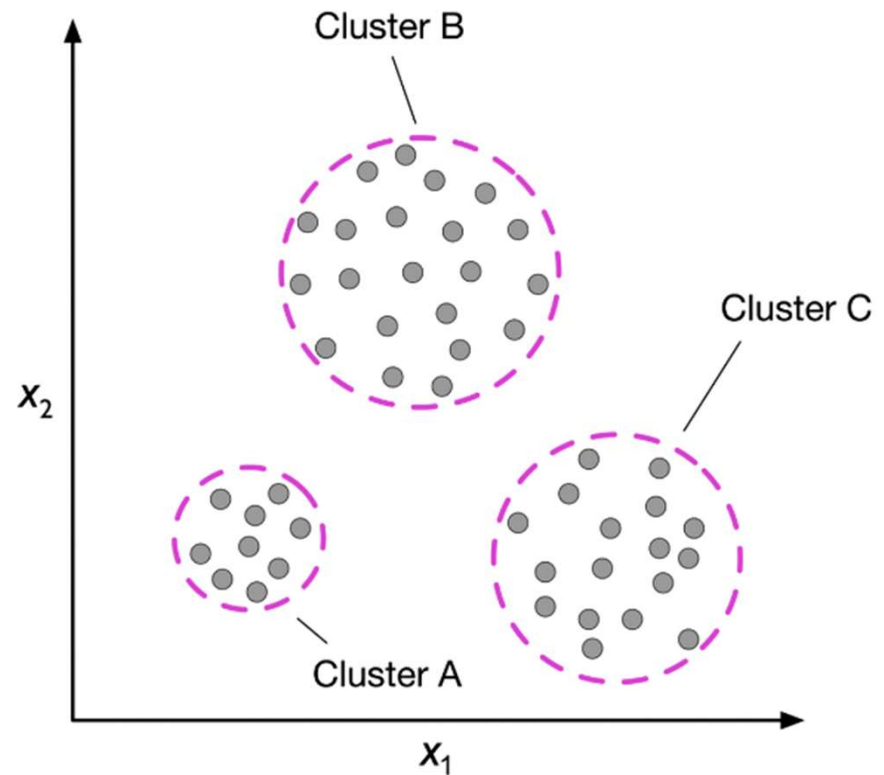


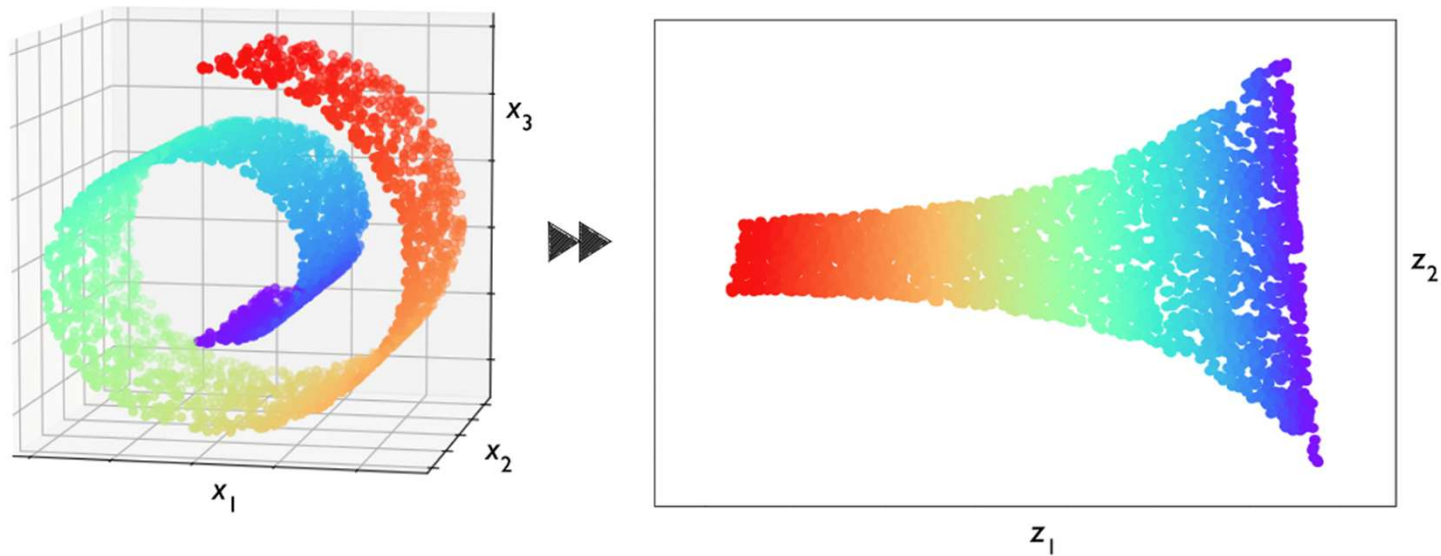
Linear regression



Unsupervised learning

Unsupervised classification : clustering



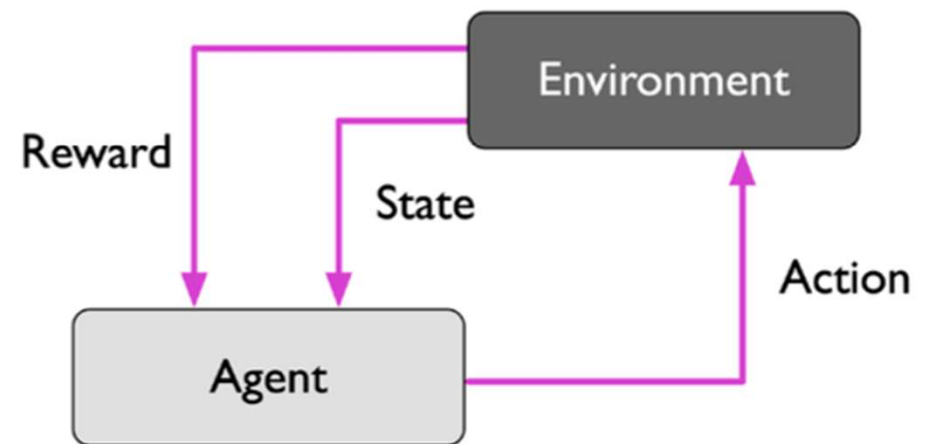


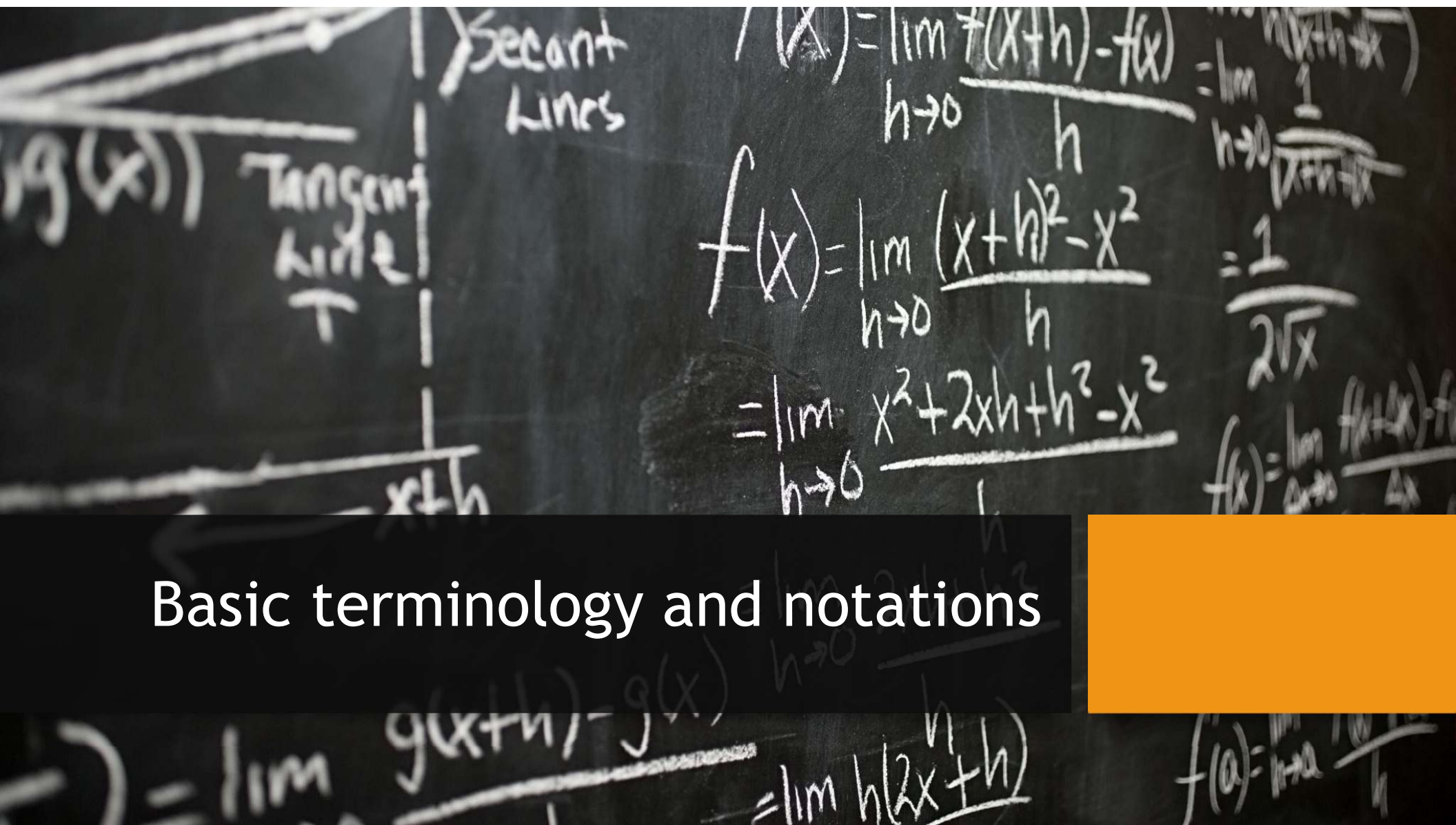
[1]

Dimensionality reduction

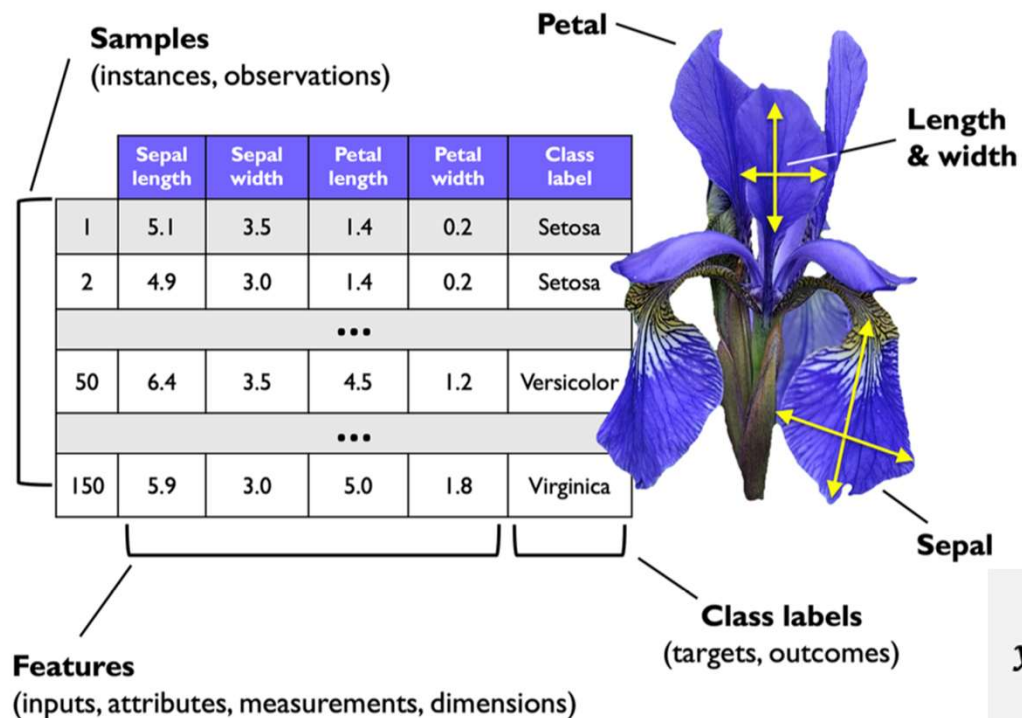
Reinforcement learning

Reinforcement learning process





Basic terminology and notations



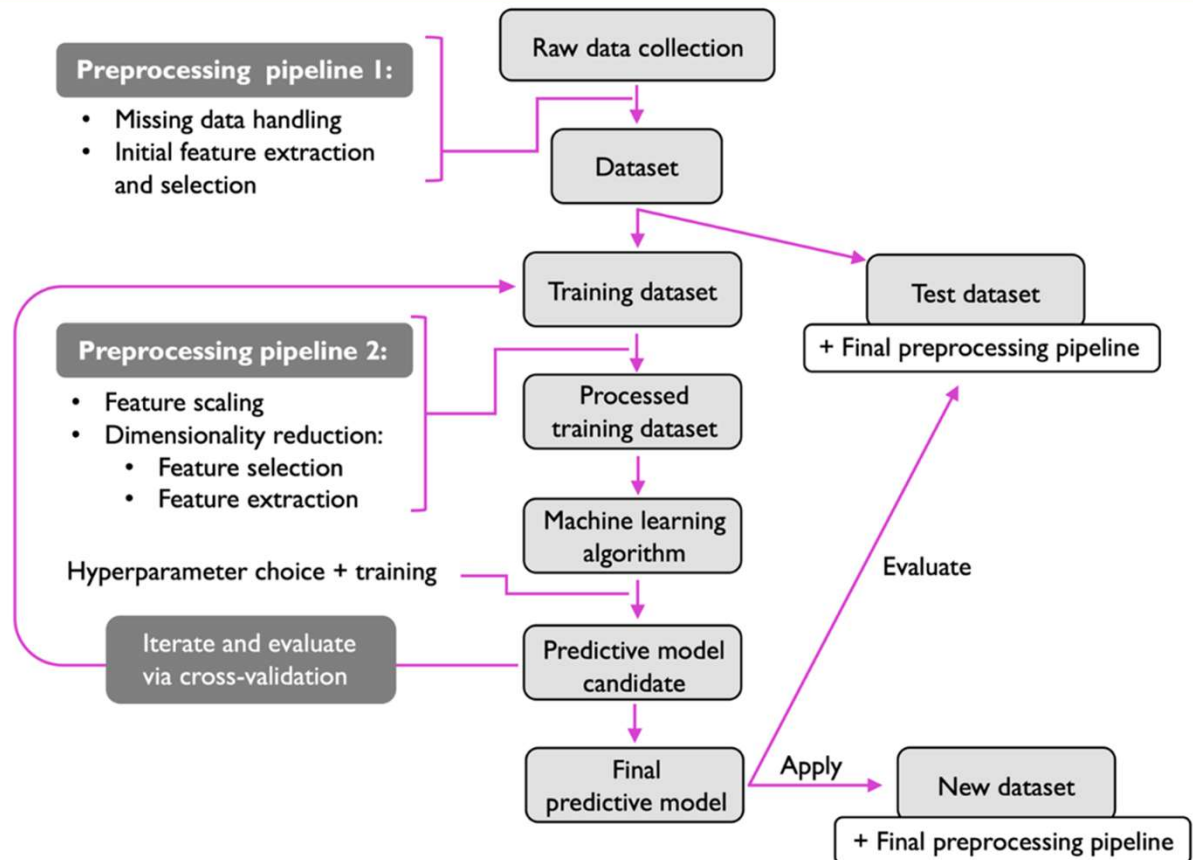
$X \in \mathbb{R}^{150 \times 4}$

$$\begin{bmatrix} x_1^{(1)} & x_2^{(1)} & x_3^{(1)} & x_4^{(1)} \\ x_1^{(2)} & x_2^{(2)} & x_3^{(2)} & x_4^{(2)} \\ \vdots & \vdots & \vdots & \vdots \\ x_1^{(150)} & x_2^{(150)} & x_3^{(150)} & x_4^{(150)} \end{bmatrix}$$

$$\mathbf{y} = \begin{bmatrix} y^{(1)} \\ \dots \\ y^{(150)} \end{bmatrix}, \text{ where } y^{(i)} \in \{\text{Setosa, Versicolor, Virginica}\}$$

Example : the iris dataset

Building machine learning system



Setup Python environment locally

Using conda

You should install anaconda or miniconda before executing the following commands on Anaconda prompt.

- `conda install -y jupyter`
- `conda create -n mlpi`
- `conda activate mlpi`
- `conda install conda-forge::jupyterlab`
- `pip install scikit-learn torch torchvision torchaudio`
- `python -m ipykernel install --user --name mlpi --display-name "mlpi"`

Then launch `jupyter lab`

Using docker

- Install docker desktop
- Create a work folder and put the provided **config.yml** and **Dockerfile** in there
- run docker desktop
- cd to that folder and type
 - docker compose up
- Open browser at local:8080
- Specify password to launch jupyter notebook

config.yml

```
services:
  notebook:
    build: ./
    ports:
      - 8080:8888
    user: root
    environment:
      - DOCKER_DEFAULT_PLATFORM=linux/amd64
      - JUPYTER_TOKEN=yourpassword
      - NB_UID=1000
      - GRANT_SUDO="yes"
    volumes:
      - ./:/home/jovyan
```


Dockerfile

```
FROM --platform=linux/amd64 quay.io/jupyter/pytorch-notebook
```

```
USER root
```

```
RUN pip install --upgrade pip && \  
    pip install scikit-learn && \  
    fix-permissions "/home/${NB_USER}"
```

Some useful online courses

- Stanford CS229 (Andrew NG : Autumn 2018)
<https://www.youtube.com/playlist?list=PLoROMvodv4rMiGQp3WXShtMGgzqpfVfbU>
- Stanford CS229 (Spring 2022)
https://www.youtube.com/playlist?list=PLoROMvodv4rNyWOpJg_Yh4NSql4Z4vOYy
- MIT 6.036 (Fall 2020)
https://www.youtube.com/playlist?list=PLxC_ffO4q_rW0bqQB80_vcQB09HOA3ClV
- R. Sebastian Intro to ML (Fall 2020)
https://youtube.com/playlist?list=PLTKMiZHVd_2KyGirGEvKlniaWeLOHhUF3&si=CDL0ERiKTeZbKO1U

References

1. S. Raschka, Y. Liu and V. Mirjalili. Machine Learning with PyTorch and Scikit-Learn. Packt Publishing. 2022.
2. F. Chollet. Deep Learning with Python 2ed. Manning Publications Co. 2021.
3. A. Zhang, Z. Lipton, M. Li and J. Smola. Dive into Deep Learning. 2022.
<https://d2l.ai/>
4. A. Ng. CS229 Lecture Notes. Stanford University. 2023.
https://cs229.stanford.edu/main_notes.pdf
5. T. Broderick. Introduction to Machine Learning course (6.036). Massachusetts Institute of Technology. 2020.
<https://tamarabroderick.com/ml.html>